

Digital Payments Transaction Analytics

Segmentation, fraud patterns & volume forecasting — synthetic mobile payment transaction data, 6.36M transactions

Findings

- 1. Fraud is concentrated in two transaction types only.** TRANSFER carries a 0.77% fraud rate and CASH_OUT 0.18%; CASH_IN, PAYMENT, and DEBIT show zero labeled fraud in this data. Fraud TRANSFERS almost always drain the account — 96% of them take the origin balance to ~0 in a single transaction, a much sharper signature than the type-level rate alone suggests.
- 2. Account value is concentrated in half the base, but it's a two-way split, not a single dominant tier.** Quantile RFM segmentation splits accounts into 5 tiers; the two high-value tiers (frequent high spenders, and high-value one-off senders) are 50% of accounts by count but hold 94.3% of total transaction value between them, roughly evenly. The other 50% of accounts — low-value-frequent, standard, and dormant — account for just 5.7% of value.

Interventions

- 1. Real-time flag on near-total balance drain.** Rather than a type-level rate threshold, flag TRANSFER transactions that take the origin account's balance to within 1% of zero in one step — the verified pattern behind 96% of labeled fraud TRANSFERS, and rare in legitimate traffic.
- 2. Tiered monitoring by RFM segment.** Light-touch checks on high-frequency regulars, whose repeat behavior gives a stable baseline to compare against; tighter scrutiny on high-value transactions from accounts with little prior activity — the profile every labeled fraud CASH_OUT in this dataset matches (each is a single transaction from an account never seen before or since).

Method & data notes

- Segmentation: quantile RFM (4 tiers on recency/frequency/monetary), not k-means — faster, interpretable, and business-ready without a training/scoring pipeline.
- Forecast: 7-day linear trend fit on the most recent week, not ARIMA/Prophet. Full-history volume shows a level shift roughly two-thirds through the window (early days run 5-14x later days); fitting on the recent regime instead of the full history avoids extrapolating the early high-volume trend into a nonsensical negative forecast.
- Checked for a literal TRANSFER-to-CASH_OUT "mule account" chain (a shared account ID linking a fraud transfer to a later cash-out) and found it's structurally absent: 0 accounts do both, and only 3 of 8,213 fraud rows share any account link at all. The two fraud types present as independent single-hop events in this file, not a traceable chain — reported here since it contradicts a common assumption about this dataset.
- Merchant accounts (nameOrig/nameDest starting "M") carry a zero-balance placeholder for oldbalanceDest/newbalanceDest — a known data artifact, excluded from balance logic rather than treated as a signal.
- Balance reconciliation (oldbalanceOrg - amount vs newbalanceOrig) mismatches on 79.8% of all rows — a data-generation characteristic of this dataset, not a fraud indicator; noted so it isn't mistaken for one downstream.